

RAG-Augmented Multi-Agent LLM Framework for Explainable Software Vulnerability Detection Using CWE Knowledge Bases

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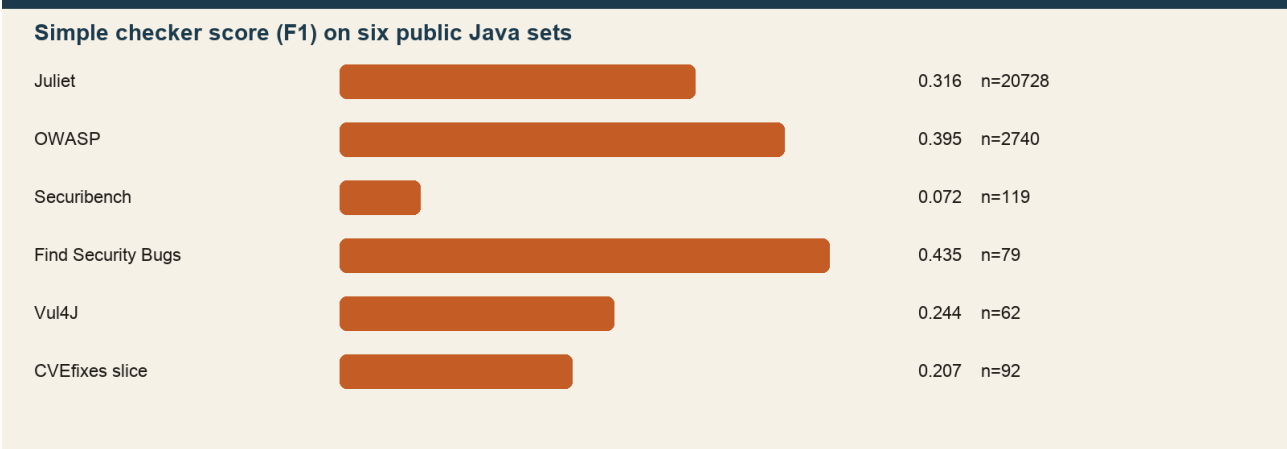
Six public Java test sets

This report adds six public test sets for the simple checker. It continues report 7.

What we did

We ran only the simple pattern checker on six named public Java sets. The scores are modest. The best F1 here is 0.435. The lowest is 0.072. A bigger set is not an easier set. We did try the live model on small samples of these sets earlier. Those scores are withdrawn. The old prompts still showed words that gave away the true label, such as a method named bad. A fair test must not show the answer inside the question.

Picture



Numbers for this step only

Test set	Files	Simple checker F1
Juliet Java	20728	0.316
OWASP Benchmark	2740	0.395
Securibench Micro	119	0.072
Find Security Bugs	79	0.435
Vul4J	62	0.244
CVEfixes Java slice	92	0.207

What is next

Next we will score the live model on Juliet pairs, after hiding the answer words.

What this report does not claim

The simple checker is not CodeQL and not a commercial scanner. The withdrawn model

scores must not be quoted.